Pharmacology, Drug Design and Synthesis

Science

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Pharmacology, Drug Design and Synthesis (A13321)

Short Title:	Pharmacol, Drug Design & Synth
Faculty	Science and Computing
Department	Science
Cluster	Pharmaceutical Science
Author	AHAYDEN
Credits	5
Level:	Advanced

Description of Module / Aims

This module covers an overview of pharmacology, with an emphasis on the relationship between chemical structure and biological activity of pharmaceuticals. The development and use of asymmetric synthetic methods for pharmaceuticals and the importance of heterocylic compounds as pharmaceuticals are included.

Programmes

PHAR-0007	BSc (Hons) in Applied Chemistry with Quality Management (WD_SQMCH_B)	4 / 2 / M
PHAR-0007	BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)	4 / 8 / M

Indicative Content

- Pharmacokinetics: principles of drug absorption, distribution, metabolism and excretion
- Relationship of ADME to routes of administration: first pass metabolism and role of blood brain barrier; bioequivalence of similar formulations
- Central role of log P, pKa and log D values in Pharmacokinetics: Lipinski's rules relating ADME to molecular structure
- Drug metabolism: phase 1 and phase 2 metabolic reactions, drug- protein binding; structure, examples and role of pro-drugs; synthetic and metabolic precursor pro-drugs
- Pharmacodynamics: receptor theory, agonists/antagonists, log dose response curves; structure and mode of action of various receptor types
- Modification of neurotransmission in pharmacodynamics: neurotransmitter agonists/antagonists/ reuptake inhibitors
- Adrenergic (sympathetic) and cholinergic (parasympathetic) neurohumoral transmission; monoamine oxidase inhibitors
- Examples of modification of neurotransmitters and related drug structures to give selective action based on receptor sub-types
- Examples of development of selective agonists/antagonists for treatment of asthma and selective cardiac drugs
- Mode of action of G protein receptor: amplification and cascade effects
- Therapeutic index: importance of ED50 and LD50 in drug safety and plasma level monitoring
- Structure/activity relationships: overview of quantitative structure/activity relationships (QSAR), isosteres and role of molecular modelling
- Case study: development and role of Tamoxifen as a selective estrogen receptor modulator; role of estrogen synthesis inhibitors
- Principles and classification of types of asymmetric synthesis of pharmaceuticals; single enantiomer and racemic drugs: role of eutomers and distomers
- Role of chiral auxiliaries, asymmetric induction, mole ratio chiral reagents and chiral catalysts; examples of asymmetric synthesis of pharmaceuticals

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Evaluate the importance of, and factors controlling, the absorption, distribution, metabolism and elimination of drugs.
2. Evaluate and explain the role of receptor theory in pharmacodynamics and its relationship to modification of neurohumoral transmission.
3. Determine the structure activity relationships for selected classes of drugs.
4. Propose examples of synthetic routes for heterocyclic pharmaceuticals.
5. Predict and evaluate asymmetric pharmaceutical synthetic methods.

Learning and Teaching Methods

- Lectures.
- Seminars.
- Self-directed study.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	100%	1,2,3,4,5

Assessment Criteria

<40%: No understanding of the basic principles of pharmacology, drug design and synthesis.

40%-49%: Ability to understand the basic principles of pharmacology, drug design and synthesis. May have some difficulty with applying the theory.

50%-59%: All the above, in addition to the ability to demonstrate an understanding of some of the complexities of the topics and predict some trends in analytical results.

60%-69%: In addition, shows a high level of competence and efficiency in the area of pharmacology, drug design and synthesis and demonstrates an ability to evaluate detailed knowledge acquired in the module to new material.

70%-100%: All previous to an excellent level. The learner will be able to demonstrate comprehensive knowledge and understanding of pharmacology, drug design and synthesis, predict general trends in behaviour, including exceptional behaviour and critically evaluate and relate topics that overlap all of the topics.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	36
Independent Learning	99	99

Essential Material

- Aitken, R.A. and S.N. Kilenyi. _Asymmetic Synthesis_. 2nd ed. London: Blackie Academic, 2000.
- Brunton, L. and B. Chabner. _The Pharmacological Basis of Therapeutics._. 12th ed. New York: McGraw and Hill, 2001.
- Krogsgaard, P. _Textbook of Drug Design and Discovery_. 3rd ed. London: Taylor & Francis, 2002.